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The e-commerce business is quite popular today. Here, you do not need to take orders by going to each customer. A company launches its website to sell the items to the end consumer, and customers can order the products that they require from the same website. Famous examples of such e-commerce companies are Amazon, Flipkart, Myntra, Paytm and Snapdeal.

Suppose you are working as a Machine Learning Engineer in an e-commerce company named 'Ebuss'. Ebuss has captured a huge market share in many fields, and it sells the products in various categories such as household essentials, books, personal care products, medicines, cosmetic items, beauty products, electrical appliances, kitchen and dining products and health care products.

With the advancement in technology, it is imperative for Ebuss to grow quickly in the e-commerce market to become a major leader in the market because it has to compete with the likes of Amazon, Flipkart, etc., which are already market leaders.

As a senior ML Engineer, you are asked to build a model that will improve the recommendations given to the users given their past reviews and ratings.



2. Building a recommendation system
3. Improving the recommendations using the sentiment analysis model
4. Deploying the end-to-end project with a user interface

- **Data sourcing and sentiment analysis**

In this task, you have to analyse product reviews after some text preprocessing steps and build an ML model to get the sentiments corresponding to the users' reviews and ratings for multiple products.

The dataset that you are going to use is inspired by this [Kaggle competition](#). We have made a subset of the original dataset, which has been provided below.

[Product Reviews Dataset](#)

This dataset consists of 30,000 reviews for more than 200 different products. The reviews and ratings are given by more than 20,000 users. Please refer to the following attribute description file to get the details about the columns of the Review Dataset.

[Product Reviews Dataset- Attribute Description](#)

The steps to be performed for the first task are given below.

1. **Exploratory data analysis**
2. **Data cleaning**
3. **Text preprocessing**



5. **Training a text classification model:** You need to build at least three ML models. You then need to analyse the performance of each of these models and choose the best model. At least three out of the following four models need to be built (Do not forget, if required, handle the class imbalance and perform hyperparameter tuning.).

1. Logistic regression
2. Random forest
3. XGBoost
4. Naive Bayes

Out of these four models, you need to select one classification model based on its performance.

- **Building a recommendation system**

As you learnt earlier, you can use the following types of recommendation systems.

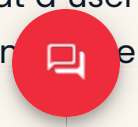
1. User-based recommendation system
2. Item-based recommendation system

Your task is to analyse the recommendation systems and select the one that is best suited in this case.

Once you get the best-suited recommendation system, the next task is to recommend 20 products that a user is most likely to purchase based on the ratings. You can use the 'reviews_username' (one of the columns in the dataset) to identify your user.

- **Improving the recommendations using the sentiment analysis model**

Now, the next task is to link this recommendation system with the sentiment analysis model that was built earlier



In this way, you will get an **ML model** (for sentiments) and the **best-suited recommendation system**. Next, you need to deploy the entire project publically.

- **Deployment of this end to end project with a user interface**

Once you get the ML model and the best-suited recommendation system, you will deploy the end-to-end project. You need to use the **Flask** framework, which is majorly used to create web applications to deploy machine learning models.

Next, you need to include the following features in the user interface.

1. Take any of the existing usernames as input.
2. Create a submit button to submit the username.
3. Once you press the submit button, it should recommend 5 products based on the entered username.

Note: An important point that you need to consider here is that the number of users and the number of products are fixed in this case study, and you are doing the sentiment analysis and building the recommendation system only for those users who have already submitted the reviews or ratings corresponding to some of the products in the dataset.

Because the dataset that you are going to use is huge, the model training may take time, and hence, you can use [Google Colab](#) to directly code or upload the already created notebook.

You can look into the user guide to access the Colab below.



models built.

What needs to be submitted for the evaluation of the project?

1. An end-to-end Jupyter Notebook, which consists of the entire code (data cleaning steps, text preprocessing, feature extraction, ML models used to build sentiment analysis models, two recommendation systems and their evaluations, etc.) of the problem statement defined
2. The following deployment files
 - One 'model.py' file, which should contain only one ML model and only one recommendation system that you have obtained from the previous steps along with the entire code to deploy the end-to-end project using Flask and Heroku
 - 'index.html' file, which includes the HTML code of the user interface
 - 'app.py' file, which is the Flask file to connect the backend ML model with the frontend HTML code
 - Supported pickle files, which have been generated while pickling the models

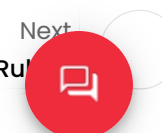
The evaluation rubrics are mentioned in the next segment.



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Evaluation Rubric



 Study Plan

Evaluation Rubric



Your solution will be evaluated based on the following rubrics.

Criteria	Meet Expectations	Does Not Meet Expectations
Task 1: Data Cleaning and Pre-Processing (10%)	Performed all data quality checks and addressed all data quality issues in the right way (especially the missing value treatment). Provided a clear explanation for missing value removal or imputation	Did not handle the missing values efficiently
	Provided a well-commented explanation for each step of data processing	Did not perform and explain the necessary steps for data processing in the comments
	Dropped the variables that were not relevant for the project with a well-commented explanation for each step. Converted all variables to correct	Did not remove unwanted variables from the columns and did not convert the variables to

	datatypes.	and not convert the variables to correct datatypes
Task 2: Text Processing (10%)	Explored all the relevant text preprocessing steps after cleaning the data set. Provided well-commented reasons for every step that was performed in text preprocessing.	Did not perform the relevant text preprocessing steps to clean the text of the reviews. Did not get clean text at the end of this step
Task 3: Feature Extraction (10 %)	Divided the data into training and testing parts Converted the text to features using the best-suited vectorizer (bag-of-words, TD-IDF, Word2Vec, etc.) for the ML model for sentiment analysis model building.	Did not divide the data into training and testing parts Did not create the appropriate vectorizer for feature extraction to build the ML model
Task 4: Model Building (20%)	Built at least 3 ML models and did the comparative analysis on why one model is better than the other three models Checked whether the data is imbalanced or not and took the necessary steps. If necessary then did the hyperparameter tuning also Selected one out of the 3 models based on performance for predicting the sentiments based on the text and title of the reviews. Provided detailed	Did not build the 3 ML models and did not give a satisfactory explanation on why one model is better than the other three models Did not consider the imbalance in the data set Did not select the ML model that performed better than the rest. Did not give reasons for

	reasons for selecting the model	selecting the model chosen
Task 5: Building the Recommendation System (20%)	Split the data set into train and test data set for the recommendation system Built at least two types of recommendation systems: user-based and item-based recommendation systems Evaluated both the types of recommendation systems and selected one based on performance. Provided detailed reasons for selecting the recommendation system	Did not split the data set into train and test datasets Did not build the two types of recommendation systems Did not evaluate the recommendation systems and did not provide any explanation for selecting the best-suited recommendation system
Task 6: Recommendation of Top 20 Products to a Specified User (10%)	Recommended the top 20 products for the username selected by the user based on the recommendation system built	Did not recommend the top 20 products for any selected username using the finalised recommendation system
Task 7: Fine-Tuning the Recommendation	Predicted the sentiment (positive or negative) of all the reviews in the train data set of the top 20 recommended products for a user. For each of	Did not predict the sentiment using the ML model of all the reviews of the top 20

System Recommendation of Top 5 Products (10%)	20 recommended products for a user. For each of the 20 products recommended, found the percentage of positive sentiments for all the reviews of each product. Filtered out the top 5 products with the highest percentage of positive reviews	recommended products for a user. Did not filter out the top 5 products based on the percentage of positive reviews
Task 8: Deployment Using Flask (10%)	Build an end-to-end web application using Flask and deploy it in your local system. You need to only include one ML model and one recommendation system to deploy the model. You need to: 1. Take the username as input, and 2. Create a submit button, and once you press the submit button, it should recommend the 5 products based on the username entered.	Was not able to deploy the project using Flask



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LJMU MS in Data Science Program – Part I

